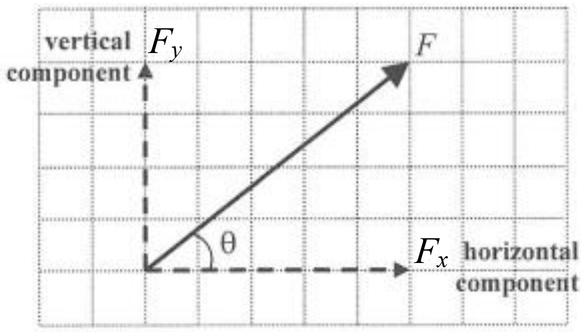
Resolution of force

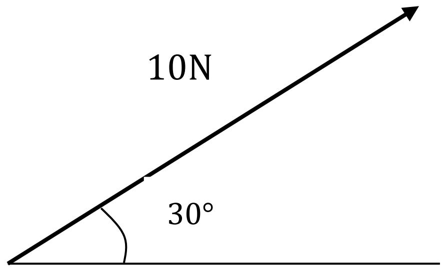
- A vector such as force can be resolved into 2 components at right angles.
- Vectors in two perpendicular directions can be considered independently.
- ***F*** showed in the graph is resolved into vertical component F_y and horizontal component F_x:

$$F_y = F\sin\theta$$

$$F_x = F\cos\theta$$

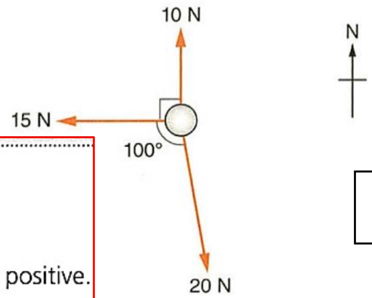


1. What is the horizontal component F_x and vertical component F_y?



F_x = (horizontal component)
= 10 cos 30°
F_y = (vertical component)
= 10 sin 30°

(For Q.2-3) Three forces act on an object as shown.



2. What is the magnitude of the net force acting on the object?

A. 5 N
B. 8.04 N
C. 15.1 N
D. 26.9 N

C

Consider the north-south direction. Take the direction towards the south as positive.
Resultant force = 20 cos (100° − 90°) − 10 = 9.696 N
Consider the east-west direction. Take the direction towards the west as positive.
Resultant force = 15 − 20 sin (100° − 90°) = 11.53 N
Net force = $\sqrt{9.696^2 + 11.53^2}$ = 15.1 N

3. What is the direction of the net force?

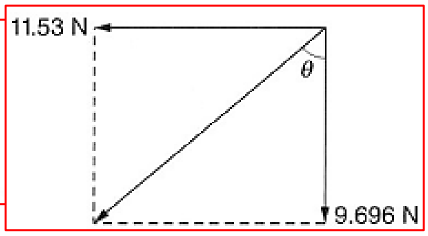
A. N35.7°E
B. N40.1°E
C. S49.9°W
D. S54.3°W

C

C $\tan\theta = \dfrac{11.53}{9.696}$

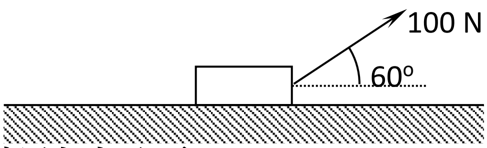
$\theta = 49.9°$

∴ The direction of the net force is S49.9°W.



4. A block of weight 100 N is placed on a smooth horizontal table. A force of 100 N at an angle of 60° is applied to the block as shown. Find the magnitude of the resultant force acting on the block.

A. 0 N
B. 50 N
C. 87 N
D. 100 N



B

5. A wooden block of mass 15 kg is pulled by a force of 80 N at an angle as shown in the figure. If the block moves at an acceleration of 2 m s^{-2}, find the friction between the block and the ground.

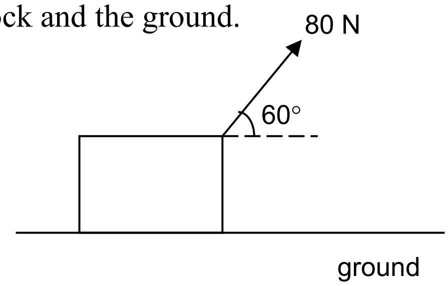
A 10 N

B 50 N

C 70 N

D 110 N

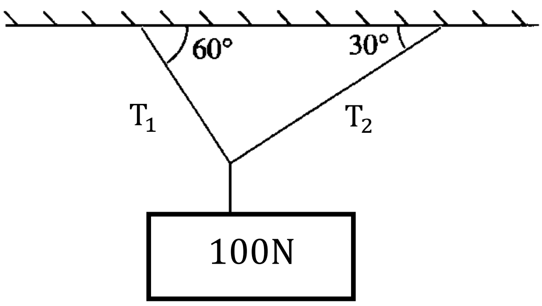
$80\cos 60° - f = 2(15)$
$f = 10N$



A

6. Find the tensions in the strings.

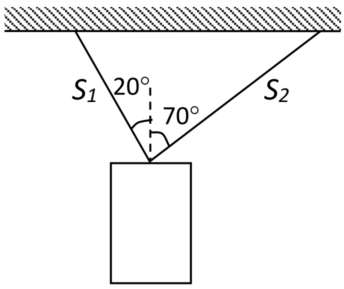
$T_1 \cos 60° = T_2 \cos 30°$..............(1)
$T_1 \sin 60° + T_2 \sin 30° = 100$......(2)
By solving (i) and (ii), we get
$T_1 = 86.6$ N
$T_2 = 50$ N



7. An object of mass 4 kg is suspended asymmetrically by two light strings S_1 and S_2 as shown below. S_1 and S_2 make angles of 20° and 70° with the vertical respectively.

(a) What is the gravitation force (weight) acting on the object?

(b) What are the tensions in the string, S_1 and S_2?



6(a) Gravitational pull = weight of object

= 39.24 N

(b) $\begin{cases} S_1 \sin 20 = S_2 \sin 70 \\ S_1 \cos 20 + S_2 \cos 70 = 40 \end{cases}$

$S_1 = 36.9$ N

$S_2 = 13.4$ N

8. 2013 DSE Q.5

A block of weight 20 N is suspended by a light string from the ceiling. A force F is applied such that the block is displaced to one side with the string making an angle 25° with the vertical as shown. Find the magnitude of F.

A. 8.5 N

B. 9.3 N

C. 18.1 N

D. 47.3 N

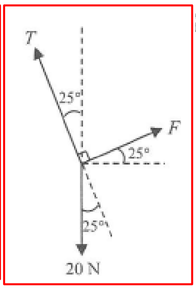
Balance of forces in x-direction : $T \sin 25° = F \cos 25°$ $\quad \therefore\ T = F\dfrac{\cos 25°}{\sin 25°}$

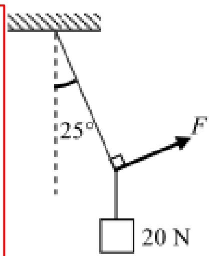
Balance of forces in y-direction : $T \cos 25° + F \sin 25° = 20$

$\therefore\ F\dfrac{\cos 25°}{\sin 25°} \times \cos 25° + F \sin 25° = 20$

$\therefore\ F\dfrac{\cos^2 25° + \sin^2 25°}{\sin 25°} = 20$

$\therefore\ F = 20 \sin 25° = 8.5$ N





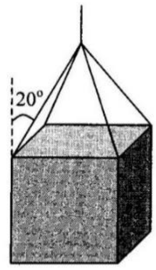
A

9. A uniform cube of weight 600 N is held in equilibrium in the air by four identical cables as shown above. If each cable makes an equal angle of 20° with the vertical, find the tension in each cable.

A. 150 N
B. 160 N
C. 412 N
D. 439 N

$T\cos 20^{o} \times 4 = 600$

$T = 159.6N$

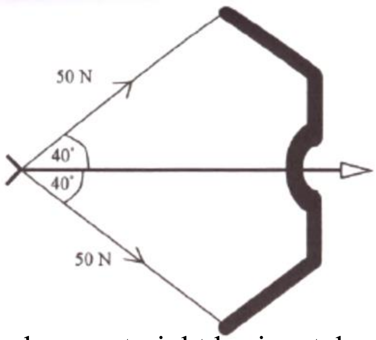


B

10. A bow is pulled horizontally to create a tension of 50 N on its string as shown in the figure. What is the initial acceleration of the arrow of mass 100 g when it is released?

A. 321 ms^{-2}
B. 383 ms^{-2}
C. 643 ms^{-2}
D. 766 ms^{-2}

Magnitude of horizontal force
= 2×50cos40° = 76.6N
Initial Acceleration
= F/m = 76.6/0.1 = 766 ms^{-2}]



D

11. A ball is suspended by a string inside a train. When the train accelerates along a straight horizontal road, the string makes an angle of 5° to the vertical. Find the acceleration of the train.

A 0.858 m s^{-2} (towards right)
B 0.858 m s^{-2} (towards left)
C 1.17 m s^{-2} (towards right)
D 1.17 m s^{-2} (towards left)

$T \sin 5° = ma$ ....(1)
$T \cos 5° = mg$ ....(2)
(1) ÷ (2),

$\tan 5° = \frac{a}{g}$

$a = g \tan 5°$

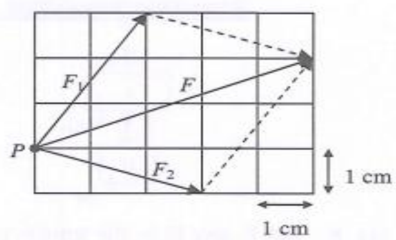
$= 9.81 \times \tan 5°$

A

12. Two forces F_1 and F_2 act on a small bead P as shown. If P is acted on by a third force F_3 in order to keep it in equilibrium, find the magnitude and direction of F_3 (scale: 1 cm represents 1 N)

	Magnitude of F_3/N	Direction of F_3
A.	5.4	↗
B.	5.4	↙
C.	6.8	↗
D.	6.8	↙

The resultant force of F_1 and F_2, says F, is shown below:



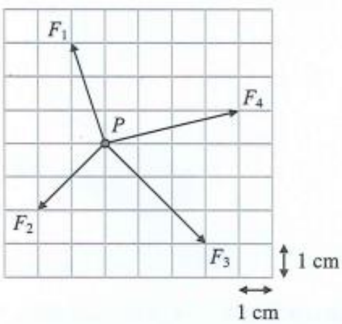
∴ Magnitude of $F = \sqrt{5^2+2^2} = 5.4$ N
Direction of F is towards upper-right.
In order to keep P in equilibrium, net force acting on it must be equal to zero. So, F_3 is equal to 5.4 N and points towards lower-left.

1 cm

B

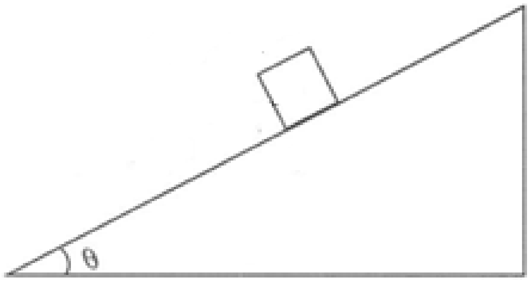
13. Particle P is subjected to four forces F_1, F_2, F_3 and F_4 as shown. Find the magnitude and direction of horizontal and vertical components of the resultant force. (scale: 1 cm represents 1 N)

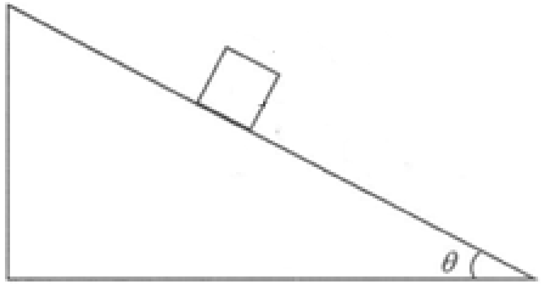
	horizontal component	vertical component
A.	8 N to the right	2 N downwards
B.	4 N to the right	2 N downwards
C.	8 N to the right	1 N downwards
D.	4 N to the right	1 N downwards



D

Resolution of force in inclined plane





Motion along a smooth inclined plane

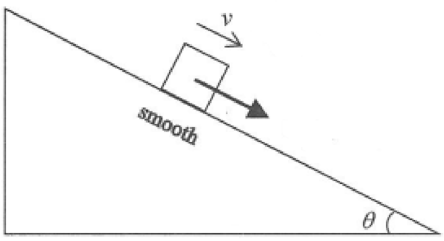
- On a smooth inclined plane, resultant force acting on the block:

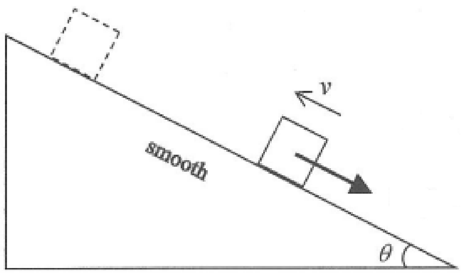
$$F_{\text{net}} = mg\ \sin\theta$$

- Acceleration of the block on a smooth incline:

$$a = g\ \sin\theta$$

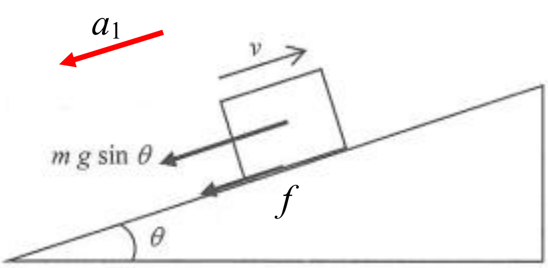
Remarks: Direction of the acceleration is always downwards along the smooth inclined plane, no matter whether the block is moving **upwards** or **downwards**. The acceleration is independent of the mass of the block on a smooth incline.



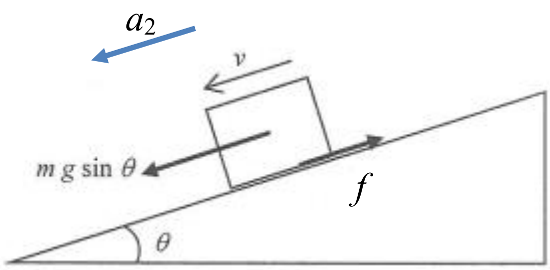


Motion along a rough inclined plane

The acceleration is **different** for upward motion and downward motion.



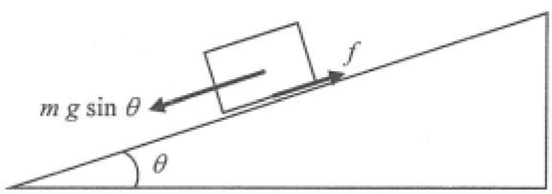
$$mg\ \sin\theta + f = ma_1$$



$$mg\ \sin\theta - f = ma_2$$

Friction-compensated

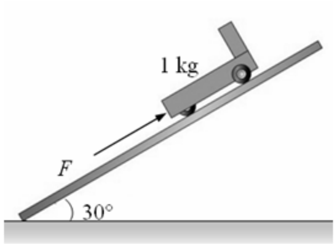
- When friction acting on an object is just balanced by the component of its weight ($mg\sin\theta$) on an inclined plane, the plane is said to be '**friction-compensated**'.



14. As shown in the figure, a 2 N force F is applied on 1 kg trolley so that the trolley is stationary? What is the value of the friction acting on the trolley F?

A. 2 N
B. 3 N
C. 5 N
D. 7 N

The system is at rest, $2 + f = 10\sin 30°$, $f = 3N$



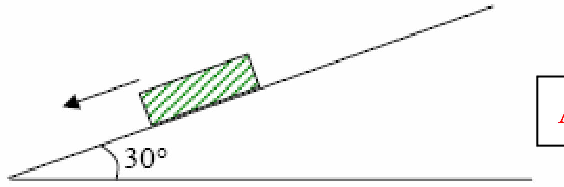
B

15. 2002 CE Q.6

A block of mass 0.5 kg slides down a rough inclined plane with an acceleration of 3 ms^{-2}. If the plane is inclined at 30° to the horizontal, find the friction between the block and the plane.

A. 1 N
B. 1.5 N
C. 2.8 N
D. 4 N

$0.5(9.81)\sin 30° - f = 3(0.5)$
$f = 0.9525N$

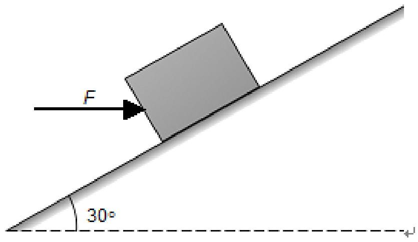


A

16. A horizontal force *F* acts on a 10 kg block as shown. It is known that the maximum friction between the block and the incline is 10 N. What is the minimum value of *F* to prevent the block from sliding down the incline?

A. 40.0 N
B. 45.1 N
C. 60.0 N
D. 80.0 N

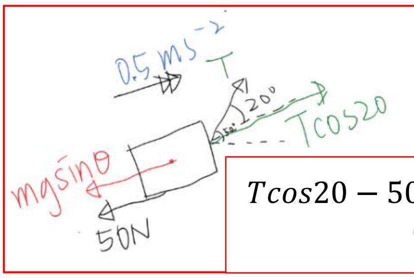
The original resultant force on the block
= 10(10)(sin30°) – 10
= 39.05 N
The component horizontal to the incline is Fcos30°.
39.05 = Fcos30°
F = 45.1 N.



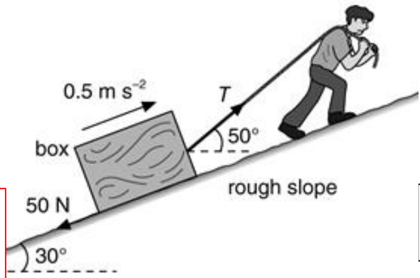
B

17. A man pulls a box of mass 40 kg and the box accelerates at 0.5 m s^{-2} up the rough slope as shown. The friction acting on the box is 50 N. The string makes an angle of 50° to the horizontal and its tension is *T*. The slope is inclined at 30° to the horizontal. Find the tension *T*.

A 74.5 N
B 283 N
C 414 N
D 778 N



$Tcos20 - 50 - mgsin\theta = 40(0.5)$
T = 283N



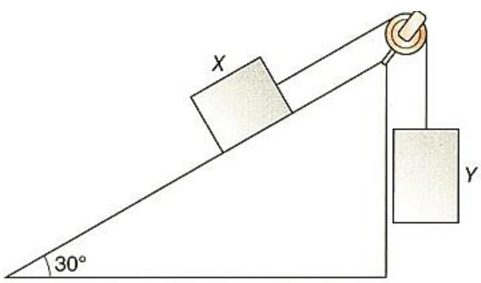
B

18. Object X on an inclined plane is connected to object Y as shown. The masses of X and Y are 1 kg and 2 kg respectively. The system remains stationary. Assume that the pulley is smooth. If the string connecting X and Y breaks suddenly, what will happen to X?

A. It remains stationary.
B. It moves down the plane with a constant velocity.
C. It moves up the plane with an acceleration of 4.91 m s^{-2}.
D. It moves down the plane with an acceleration of 4.91 m s^{-2}.

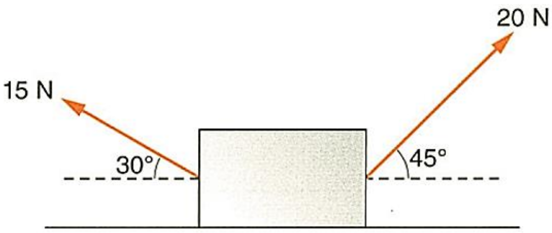
A Let the friction between X and the plane be f. Initially,
$m_Y g = m_X g \sin\theta + f$
$f = m_Y g - m_X g \sin\theta = 2 \times 9.81 - 1 \times 9.81 \sin 30° = 14.7$ N

After the string breaks,
downward force on X along the plane $= m_X g \sin\theta = 4.91$ N < 14.7 N
∴ X remains stationary.



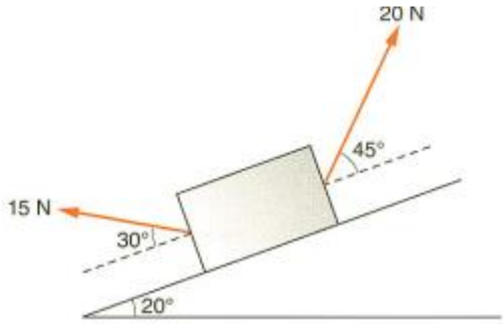
A

19. Two forces are acting on a 5 kg block on a smooth horizontal plane as shown.
    (a) Find the normal force acting on the block by the plane.
    (b) Find the magnitude and direction of the acceleration of the block.
    (c) The block is now put on a smooth slope as shown. Find the normal force acting on the block.
    Also, find the magnitude and direction of the acceleration of the block.



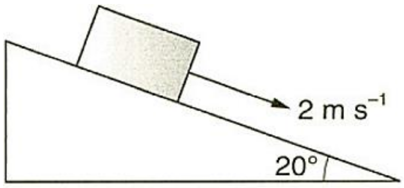
(a) Take the upward direction as positive.
$R + 15 \sin 30° + 20 \sin 45° = mg$ 1M
$R = 27.4$ N 1A
∴ The normal force is 27.4 N.

(b) Take the direction to the right as positive.
By $F = ma$, 1M
$20 \cos 45° - 15 \cos 30° = ma$
$a = 0.230$ m s^{-2} 1A
∴ The acceleration is 0.230 m s^{-2} towards the righ

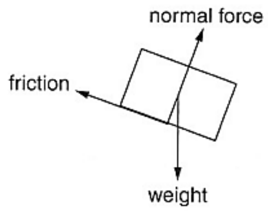


(c) In the direction perpendicular to the slope,
$R + 15 \sin 30° + 20 \sin 45° = mg \cos 20°$ 1M
$R = 24.4$ N 1A
∴ The normal force is 24.4 N.
Take the direction down the slope as positive.
$ma = 15 \cos 30° + mg \sin 20° - 20 \cos 45°$ 1M
$a = 3.12$ m s^{-2} 1A
∴ The acceleration is 3.12 m s^{-2} down the slope. 1A

20. A block of mass 3 kg moves down an inclined plane at a constant velocity of 2 m s^{-1} as shown.
    (a) Draw a free-body diagram for the block.
    (b) Find the magnitudes of all the forces shown in **(a)**.



(b) Weight = mg = 3 × 9.81 = 29.4 N
Since the block moves at a constant velocity, no net force acts on it.
Consider the direction parallel to the plane.
Friction = $mg \sin \theta$ = 3 × 9.81 sin 20° = 10.1 N
Consider the direction perpendicular to the plane.
Normal force = $mg \cos \theta$ = 3 × 9.81 cos 20° = 27.7 N

(a)


(1 correct force with correct name)
(All correct)

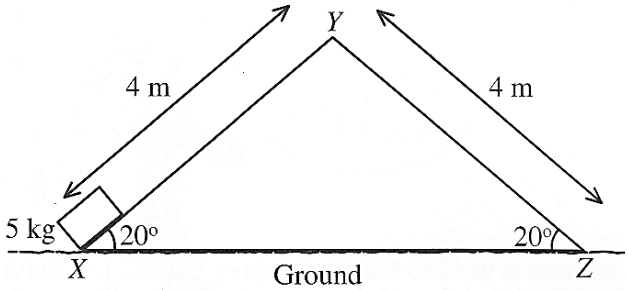
21. A triangular block XYZ is fixed on the ground. A 5 kg rectangular block moves up the triangular block from X with initial speed u. The friction between the two blocks has a constant value 12 N.

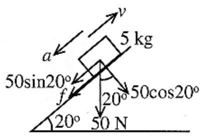
(a) Find the acceleration of the block when it slides up plane XY.

(b) The block slides across the triangular block with minimum initial speed u.

(i) Find the value of u.

(ii) Find the speed of the block when it arrives Z.





The free body diagram of the block when sliding up is shown above.

$F = ma$

$mg\sin 20^\circ + f = ma$

$a = g\sin 20^\circ + \dfrac{f}{m}$

$a = 10\sin 20^\circ + \dfrac{12}{5} = 5.82 \text{ m s}^{-2}$

The acceleration points downward along the plane.

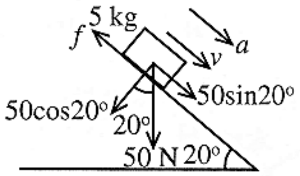
(i) With the minimum u, the block becomes instantaneous at rest on the top of Y.

$v^2 = u^2 + 2as$

$0^2 = u^2 + 2 \times (-5.82) \times 4$

$u = 6.82 \text{ m s}^{-1}$

(ii)



When sliding down along YZ,

$mg\sin 20^\circ - f = ma$

$a = g\sin 20^\circ - \dfrac{f}{m}$

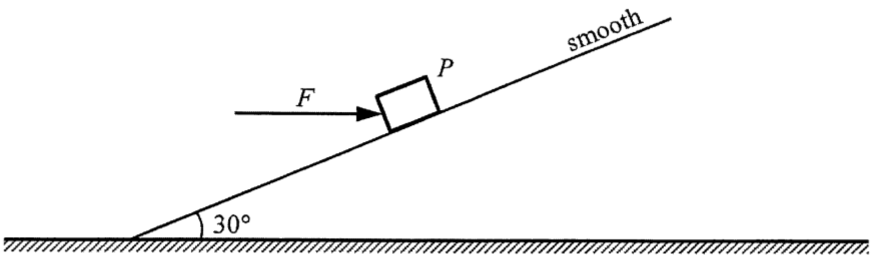
$a = 10\sin 20^\circ - \dfrac{12}{5} = 1.02 \text{ m s}^{-2}$

$v^2 = u^2 + 2as$

$v^2 = 0^2 + 2 \times 1.02 \times 4$

$v = 2.86 \text{ m s}^{-1}$

22.



(a) A block P of mass 10 kg is kept stationary on a smooth incline by a horizontal force F as shown in Figure 5.1. The incline makes an angle of 30° with the horizontal. ($g = 9.81$ m s^{-2})

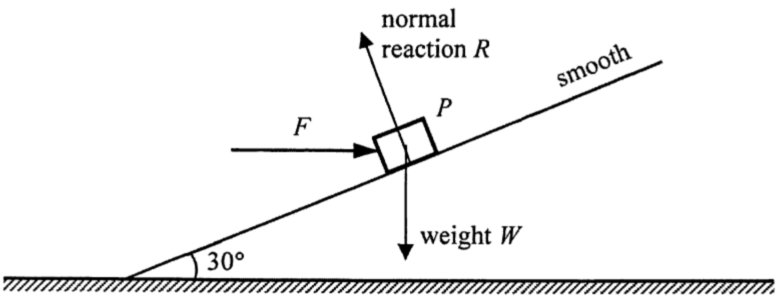
(i) On Figure 5.1, indicate and label all other forces acting on P. (2 marks)

(ii) Find the magnitudes of the force F and the force exerted by the block on the incline respectively. (3 marks)

(b) Now F is removed and neglect air resistance.

(i) What is the magnitude of the acceleration of the block ? (1 mark)

(ii) Explain whether the force exerted by the block on the incline would increase, decrease or remain unchanged when compared with that in (a)(ii). (2 marks)

(a) (i) 	1A 1A	Forces R(or N) and W(or Mg) correctly indicated and labelled.
	2	
(ii) $R\cos 30° = W = Mg$ and $R\sin 30° = F$	1M	$M = 10$ kg; $Mg = 98.1$ N
$F = Mg \times \tan 30° = 56.63806\text{ N} \approx 56.6\text{ N}$	1A	For $g = 10$ m s^{-2}, 57.735027 N ≈ 57.7 N
$N = R = \dfrac{Mg}{\cos 30°} = 113.27612\text{ N} \approx 113\text{ N}$	1A	115.470054 N ≈ 115 N
Or $R = W\cos 30° + F\sin 30°$ and $W\sin 30° = F\cos 30°$	1M	
	3	
(b) (i) $g\sin\theta = 9.81\sin 30° = 4.905\text{ m s}^{-2} \approx 4.91\text{ m s}^{-2}$	1A	5 m s^{-2} for $g = 10$ m s^{-2}
	1	
(ii) Decrease	1A	
as the component of F perpendicular to the incline no longer acts on the block / incline	1A	
Or as F is removed, the force pressing the block / incline would decrease (only the weight's component left)	1A	
	2	